

Oxford Cambridge and RSA Examinations

GCE

Economics

H460/03: Themes in economics

A Level

Predicted Paper — ADVANCED

Mark Scheme

SECTION A — Multiple Choice Answers

Question	Answer	Question	Answer
1	C	16	A
2	B	17	D
3	B	18	B
4	B	19	C
5	A	20	B
6	C	21	A
7	C	22	C
8	B	23	D
9	B	24	C
10	B	25	B
11	B	26	A
12	B	27	B
13	C	28	A
14	C	29	B
15	D	30	B

Section A — Full Rationales

Q	Key	Rationale	AO
1	C	Wages paid for labour are a factor reward. Unemployment benefit is a transfer payment — the recipient is not providing a factor service in exchange.	AO1
2	B	With MPC = 0.75 out of disposable income and tax rate $t = 20\%$: effective MPC out of gross income = $0.75 \times (1 - 0.20) = 0.60$. Multiplier $k = 1/(1 - 0.60) = 2.5$. $\Delta Y = £100bn \times 2.5 = £250bn$.	AO2
3	B	In a liquidity trap, a small (0.25pp) cut in the base rate is largely ineffective because the zero-lower-bound means further rate cuts cannot reach households via lending channels. Unconventional policy (QE, forward guidance) and fiscal stimulus remain effective.	AO3
4	B	Selling bonds is CONTRACTIONARY. Bond prices fall → yields rise; bank reserves fall → lending falls → AD falls. Options A, C, D describe expansionary policy (the opposite).	AO3
5	A	Joint supply (complements in PRODUCTION): rising demand for Y raises its price, incentivising producers to produce more Y — and automatically more X as a by-product. Supply of X shifts right → price of X falls.	AO3
6	C	Cournot duopoly with $P = 100 - Q$ and $MC = 10$: $q_i^* = (a - c)/((n+1)b) = (100 - 10)/(3 \times 1) = 30$. Each firm produces 30; $Q = 60$; $P = 100 - 60 = £40$.	AO3
7	C	Network effects create positive feedback: the bigger the platform, the more valuable to join → market tips toward a dominant firm. This is the 'winner-takes-most' dynamic characteristic of digital markets.	AO1
8	B	$P = 120 - 2Q \rightarrow TR = 120Q - 2Q^2 \rightarrow MR = 120 - 4Q$. $MC = dTC/dQ = 20$. $MR = MC: 120 - 4Q = 20 \rightarrow Q = 25$. $P = 120 - 50 = £70$. $TR = 1,750$. $TC = 20 + 500 = 520$. Profit = 1,230.	AO2
9	B	$MU = dTU/dQ = 10 - Q$. $MU(1) = 9$; $MU(2) = 8 \rightarrow$ the 2nd unit is the first unit where additional utility falls below the previous. TU maximised when $MU = 0 \rightarrow Q = 10$.	AO2
10	B	Forward cost: $\$1,000,000 / 1.20 = £833,333$. Spot in 90d: $\$1,000,000 / 1.30 = £769,231$. The forward contract cost MORE (by £64,103) because the £ actually appreciated to \$1.30 (the firm locked in a worse rate).	AO2
11	B	Inverse-elasticity rule: $(P - MC)/P = 1/ PED $. Market A: $1/2 = 0.50$. Market B: $1/4 = 0.25$. Ratio A:B = 2:1. Mark-up is higher in the less-elastic market, as expected.	AO3
12	B	$80,000 \times (1.02)^t = 50,000 \times (1.04)^t \rightarrow (1.04/1.02)^t = 1.6 \rightarrow t = \ln(1.6)/\ln(1.01961) = 0.47000/0.01942 \approx 24.2$ years.	AO2
13	C	Set $MCL = MRP: 10 + L = 70 - L \rightarrow 2L = 60 \rightarrow L = 30$. Wage from the supply curve at $L = 30$: $W = 10 + 0.5(30) = £25$. Monopsony wage (£25) < MRP at $L=30$ (£40) — the classic monopsonistic 'wage gap'.	AO3
14	C	$MV = PY \rightarrow \% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y$. With stable V ($\% \Delta V = 0$): $8\% + 0\% = \% \Delta P + 3\% \rightarrow$ inflation = 5%.	AO2
15	D	Marshall-Lerner: depreciation improves CA only if $ PED_X + PED_M > 1$. Here $0.3 + 0.4 = 0.7 < 1 \rightarrow$ condition fails → CA worsens in the short run (this is also the J-curve's starting point).	AO3
16	A	P: tax = $(20,000 - 15,000) \times 0.25 = £1,250$; avg rate = $1,250/20,000 = 6.25\%$. Q: tax = $(80,000 - 15,000) \times 0.25 = £16,250$; avg rate = $16,250/80,000 = 20.3\%$. Higher earner pays higher AVERAGE rate → tax is progressive despite flat marginal rate.	AO2
17	D	The Gini coefficient is the area between the Lorenz curve and the 45° line. If two Lorenz curves CROSS, one can be closer to equality over part of the	AO3

		distribution and further over another — making Gini rankings ambiguous.	
18	B	Incidence falls on the side of the market that is LESS elastic. Here demand is highly inelastic — consumers bear most of the burden. Because Q falls little, government revenue ($t \times Q$) is large.	AO3
19	C	Injections = $I + G + X = 180 + 220 + 400 = 800$. Leakages = $S + T + M = 150 + 200 + 450 = 800$. Injections = leakages → the economy is in equilibrium.	AO2
20	B	Akerlof's 'lemons' model: asymmetric information causes buyers to offer average-quality prices. Sellers of high-quality cars withdraw; only low-quality 'lemons' remain. The market unravels — classic adverse selection.	AO1
21	A	A lump-sum tax doesn't affect MC, so short-run output is unchanged. But AC rises → firms below normal profit exit in the long run → industry supply shifts left → price rises until survivors earn normal profit. Industry output is lower; fewer firms.	AO3
22	C	Self-preferencing leverages dominance from the upstream market (the platform/infrastructure) into a downstream market (the products sold via the platform). This is vertical foreclosure — the competition-law concept directly applied in DMA/CMA enforcement.	AO3
23	D	A SMALL open economy cannot affect world prices — so it cannot improve its terms of trade via a tariff. (This is the crucial qualification distinguishing small vs large open economies in trade theory.) The other three effects are all present.	AO3
24	C	£50bn of new reserves; banks hold 30% (£15bn) as excess reserves and lend out £35bn. The £35bn then multiplies through the banking system at the money multiplier of 10: $£35bn \times 10 = £350bn$ of new broad money.	AO2
25	B	A binding minimum price creates excess supply (A ✓), causes DWL (C ✓) and reduces CS (D ✓). Whether PS rises is ambiguous: though price rises, quantity sold falls; with costly unsold output, PS can fall. Hence 'unambiguously rises' is NOT a consequence.	AO3
26	A	Perfect contestability + zero sunk costs force prices to AC — because any supernormal profit would invite hit-and-run entry. With economies of scale, the firm operates at minimum AC (= normal profit). This is the key result of contestability theory.	AO1
27	B	Negative SRAS shock: SRAS shifts left → $P \uparrow$, $Y \downarrow$. Expansionary fiscal: AD shifts right → $P \uparrow$, $Y \uparrow$. Combined: P rises unambiguously (both effects push up), but Y is ambiguous (effects offset).	AO3
28	A	$HDI_W = (0.90 \times 0.85 \times 0.95)^{1/3} = (0.72675)^{1/3} \approx 0.8992$. $HDI_V = (0.95 \times 0.95 \times 0.75)^{1/3} = (0.676875)^{1/3} \approx 0.8775$. W is higher despite lower average, because geometric mean penalises imbalance — V's weak income pillar drags it down.	AO2
29	B	Lewis two-sector model: unlimited supply of surplus labour from traditional (rural) sector keeps urban wages pegged slightly above subsistence. The resulting profits are reinvested, driving capital accumulation and industrial growth until the Lewis turning point.	AO1
30	B	Primary income = factor income to/from abroad (compensation of employees, investment income). Dividends received by a UK resident from a French company = CREDIT on UK primary income. A, D are imports (debits on goods/services). C is secondary income (debit).	AO1

SECTION B — Guidance

Question 31 (2 marks)

Using the data in Fig. 1.1, calculate the percentage increase in the combined market capitalisation of the five Big Tech firms between 2019 and 2024.

Award 2 marks for correct answer:

- Combined 2019: \$4.93 trillion. Combined 2024: \$11.85 trillion.
- % increase = $(11.85 - 4.93) / 4.93 \times 100 = 140.4\%$.

Award 1 mark for:

- Correct method with arithmetic error, OR the absolute change (+\$6.92 trillion) without % conversion.

Question 32 (3 marks)

Explain, using examples from Extract 1, two reasons why digital markets tend towards 'winner-takes-most' outcomes.

Award 1 mark per identified reason (max 2); 1 mark for application to Extract 1.

- Network effects: the more users a platform has, the more valuable it is — e.g. Meta's 80%+ weekly UK reach is self-reinforcing.
- Economies of scale / high fixed costs of software development: large firms spread R&D costs over huge user bases — e.g. Google's 91% search share reflects scale advantages.
- Economies of scope from data: data gathered from one service improves others (e.g. Alphabet leveraging search data into ads, maps, YouTube).
- High switching costs / ecosystem lock-in: the Apple App Store and Google Play together handle 99% of UK downloads.

Question 33* (15 marks)

Evaluate, using an appropriate diagram and the information in Extract 1, whether increased regulation of Big Tech firms is likely to improve economic welfare.

Level / Mark	Descriptor
Level 3 (11–15 marks)	Good knowledge and understanding of relevant economic ideas, principles and models. Good application of economic concepts to the context and scope of the question, or good use of data where appropriate. Strong analysis characterised by consistently well-developed chains of reasoning; relevant diagram(s) accurate and integrated. Strong evaluation characterised by well-developed chains of reasoning; a logical and well-supported judgement.
Level 2 (6–10 marks)	Good knowledge and understanding of relevant economic ideas. Reasonable application of economic concepts, possibly inconsistent. Reasonable analysis with undeveloped chains of reasoning; diagrams may contain errors or not be integrated. Reasonable evaluation with some developed chains; an unsupported judgement.
Level 1 (1–5 marks)	Reasonable knowledge and understanding of relevant economic ideas.

	Limited application; data use may be generic or irrelevant. Limited analysis with unclear structure; simple cause/consequence statements; diagram errors that affect validity. Limited evaluation with simple statements; stated judgement or absent.
0 marks	Response is not worthy of credit

Indicative Content

Knowledge:

- Monopoly power; economic welfare (consumer + producer surplus); market failure; network effects; dynamic efficiency; regulatory failure; vertical foreclosure; price discrimination.

Application:

- Google 91% UK search share; 99% app-store concentration; Meta 80%+ weekly UK reach.
- UK Digital Markets Act 2024 — SMS designations; fines up to 10% of global turnover.
- Big Tech R&D ratios vastly exceed market averages (Meta 27%, Alphabet 14% vs S&P average 4%).
- App-store fees of 15–30% on developers.

Analysis (regulation improves welfare):

- Standard monopoly diagram: monopolist restricts output below allocatively efficient level ($P > MC$) → deadweight welfare loss. Shade triangle between AR and MC curves from Q_{mon} to Q_{comp} .
- Requiring Apple to open payment systems or Google to de-bias search rankings directly addresses vertical foreclosure — restoring competition in downstream markets (app payments, shopping comparison).
- Reducing platform fees (currently 15–30%) passes through to smaller app developers and to consumers.
- Consumer 'cost' through data collection is under-priced; regulation on data portability/privacy internalises this negative externality.

Evaluation (regulation may reduce welfare):

- Dynamic efficiency: Big Tech R&D (14–27% of revenue) dwarfs market averages; regulation could deter investment and slow innovations like generative AI. A Schumpeterian argument: temporary monopoly profit funds socially valuable R&D.
- Network effects deliver consumer surplus: each user benefits from large, integrated ecosystems. Breaking them up can destroy value.
- Regulatory failure: the CMA may have information asymmetries with tech firms; definitions like 'Strategic Market Status' risk being arbitrary; regulatory capture possible.
- Digital markets are still contestable at the frontier: entry by new AI-native firms (OpenAI, Anthropic, Mistral) has rapidly challenged incumbents — suggests the 'monopoly' is temporary.
- Extra-territorial effects: UK/EU regulation of global firms may cause service withdrawal (as when Meta threatened to exit EU over data rules) — welfare loss for consumers.

Judgements might include:

- Depends on the type of regulation: ex-ante conduct rules (transparency, interoperability) may be lower-cost than ex-post break-up.
- Depends on the pace of technological change and contestability of the market (AI entrants suggest high contestability at the frontier).

- Depends on whether regulators can credibly commit to targeted, proportionate action rather than capture.
- Depends on whether 'welfare' is defined in static (price) or dynamic (innovation) terms.

Question 34 (3 marks)

Using the data in Fig. 2.1 and Fig. 2.2, calculate (a) the percentage change in India's HDI and (b) the percentage change in India's GDP per capita (PPP) between 1990 and 2022.

Award 1 mark for each correct calculation (max 2); 1 mark for correct presentation of both with % sign.

- (a) HDI change: $(0.644 - 0.434) / 0.434 \times 100 = 48.4\%$.
- (b) GDP per capita change: $(8,790 - 1,180) / 1,180 \times 100 = 645\%$.

Accept answers rounded to the nearest whole percent (48%, 645%).

- Answer highlights that GDP per capita grew much faster than HDI — the development gap is bigger in money terms than in human welfare terms.

Question 35 (2 marks)

Using Fig. 2.1, identify TWO limitations of using HDI alone to compare living standards across the three countries.

Award 1 mark per valid limitation (max 2).

- HDI does not capture income inequality — India's Gini is 0.65 while Vietnam's is 0.36, yet HDI averages hide this.
- HDI ignores environmental sustainability / pollution.
- HDI omits political freedoms, gender equality, and human rights.
- HDI doesn't reveal the distribution of health and education outcomes within a country.
- HDI uses averages, masking regional disparities (rural-urban).

Question 36* (15 marks)

Evaluate, using the information in Extract 2, whether rapid economic growth is the most effective route to raising living standards in emerging economies such as India and Vietnam.

Level / Mark	Descriptor
Level 3 (11–15 marks)	Good knowledge and understanding of relevant economic ideas, principles and models. Good application of economic concepts to the context and scope of the question, or good use of data where appropriate. Strong analysis characterised by consistently well-developed chains of reasoning; relevant diagram(s) accurate and integrated. Strong evaluation characterised by well-developed chains of reasoning; a logical and well-supported judgement.
Level 2 (6–10 marks)	Good knowledge and understanding of relevant economic ideas. Reasonable application of economic concepts, possibly inconsistent. Reasonable analysis with undeveloped chains of reasoning; diagrams may contain errors or not be integrated. Reasonable evaluation with some developed chains; an unsupported judgement.
Level 1 (1–5 marks)	Reasonable knowledge and understanding of relevant economic ideas. Limited application; data use may be generic or irrelevant. Limited analysis with unclear structure; simple cause/consequence statements; diagram errors that affect validity. Limited evaluation with simple statements; stated judgement or absent.

0 marks	Response is not worthy of credit
---------	----------------------------------

Indicative Content

Knowledge:

- Economic growth vs development; HDI vs GDP; Lewis dual-sector model; poverty; inequality; middle-income trap; FDI; human capital.

Application:

- India's real GDP growth $\approx 6.2\%$; Vietnam $\approx 6.5\%$ over three decades.
- India HDI from 0.434 to 0.644; Vietnam from 0.482 to 0.726 (now 'high').
- Structural shift: India's agricultural employment share from 60% to 44%; Vietnam 70% to 27% — Lewis-style labour reallocation.
- India's wealth Gini rose from 0.54 to 0.65; top 1% captures 22% of income.
- 230 million Indians below national poverty line; Vietnam 8% of GDP in FDI inflows.

Analysis (growth raises living standards):

- Higher GDP per capita funds private consumption, better nutrition, housing. Real-wage gains as workers move from agriculture to higher-productivity sectors (Lewis model in action).
- Higher tax revenue from growth funds public provision of healthcare, education and infrastructure — directly raising HDI components.
- Growth attracts FDI (Vietnam's 8%-of-GDP inflow) which transfers technology and raises productivity — positive feedback loop.
- Growth reduces absolute poverty: Vietnam's HDI rise from 0.482 to 0.726 correlates with growth-driven gains in life expectancy and schooling.

Evaluation (growth alone is insufficient or harmful):

- Distributional failure: India's wealth Gini ROSE to 0.65 — rapid growth has been highly unequal, with 230m still in poverty. HDI gains 0.434 $\rightarrow$ 0.644 much smaller proportionally than GDP per capita's 5-fold rise.
- Environmental degradation and pollution impose welfare losses not captured in GDP — urban air quality in India; deforestation in Vietnam.
- Middle-income trap: cheap-labour advantage erodes before innovation capacity is built — growth can stall at middle-income levels.
- Targeted social policy (Kerala model, Vietnam's universal primary education) can raise HDI even at modest growth rates — direct public spending on health/education more efficient per \$ than market-led growth.
- Quality of jobs matters: informal/gig employment may increase alongside GDP but offers no security, raising inequality and insecurity.

Judgements might include:

- Depends on whether growth is inclusive — Vietnam (low Gini) shows broader HDI gains; India (high Gini) shows weaker HDI gains despite similar growth rates.
- Depends on complementary policies: progressive taxation, public education, labour protections.
- Depends on time horizon — in the short run, growth may hurt the poor (displacement, pollution); in the long run, it may lift them.
- Growth is necessary but not sufficient — development economics increasingly favours broader metrics (HDI, multidimensional poverty index).

Question 37 (2 marks)

Using Fig. 3.1, compare the change in median net hourly pay for gig workers with the change in the NLW between 2016 and 2024.

Award 1 mark for the similarity/directional comparison; 1 mark for supporting data.

- Gig pay fell: £7.40 → £6.20 (a fall of about £1.20 or –16%).
- NLW rose: £7.20 → £11.44 (a rise of about £4.24 or +59%).
- The gap has reversed: in 2016 gig pay was slightly above NLW; by 2024 gig pay is substantially below NLW (–£5.24 gap).

Only credit answers that compare both series and cover the full period.

Question 38 (8 marks)

Evaluate, using the information in Extract 3, whether gig-economy workers should be reclassified as full employees entitled to the National Living Wage, paid holiday and employer pension contributions.

Level / Mark	Descriptor
Level 2 (5–8 marks)	Good knowledge and understanding of relevant economic ideas. Good application of economic concepts to the context; good use of data. Strong analysis with well-developed chains of reasoning; any relevant diagrams accurate and integrated. Strong evaluation with well-developed chains; a logical, well-supported judgement.
Level 1 (1–4 marks)	Reasonable knowledge and understanding of relevant economic ideas. Reasonable application, possibly inconsistent. Reasonable analysis in the form of single links. Limited evaluation with simple statements; stated or absent judgement.
0 marks	Response is not worthy of credit

Indicative Content

Knowledge: labour markets; monopsony; minimum wages; incidence of employment regulation; employment protection and elasticity of labour demand; worker welfare vs labour market flexibility.

Application:

- 4.7m UK workers (14% of workforce); gig pay £6.20/hr vs NLW £11.44.
- Uber ruling (2021) raised Uber's UK costs ≈15%.
- Algorithmic monopsony: 70%+ of income from single platform; switching deterred.
- 62% cite flexibility as main reason to take gig work.

Analysis (for reclassification):

- Monopsony analysis: platforms set wages below MRP. Reclassification + NLW raises wage closer to competitive level AND may raise employment (classic monopsony response to minimum wages).
- Equity: £6.20/hr is below NLW and below poverty wage. Treating gig workers differently creates a parallel, underpaid labour force — violates horizontal equity.
- Reduces moral hazard for platforms: current classification externalises costs (illness, old age) onto the state (NHS, UC benefits).

- Supports the 2021 Supreme Court's reasoning: substance over form — drivers ARE dependent on Uber.

Evaluation (against reclassification):

- Incidence: 15% cost rise may be passed through in higher fares → consumers bear the cost. Lower demand for rides → fewer gig jobs.
- Flexibility lost: 62% value flexibility; requiring employee status can force platforms into fixed shifts, harming students/carers.
- Job losses: platforms may reduce driver numbers and raise selection bars; existing informal workers pushed out.
- Alternative tools: sectoral minimum hourly rates, pension auto-enrolment, or a new 'dependent contractor' category may capture benefits without eliminating flexibility.

Judgements might include:

- Depends on the strength of algorithmic monopsony power and the wage gap vs NLW — evidence of a £5+ gap in 2024 is substantial.
- Depends on design: full employee status vs narrower 'worker' rights (as in the UK 2021 ruling).
- Depends on the demand elasticity for platform services and the share of costs passed to consumers vs absorbed by platforms.
- A graduated approach (NLW floor + flexibility preserved) likely delivers the best welfare balance.

Assessment Objectives Grid

Question	AO1	AO2	AO3	AO4	Total	QS
Section A Total	13	9	8	—	30	(15)
31	—	2	—	—	2	(2)
32	1	2	—	—	3	
33*	2	3	4	6	15	
34	—	2	1	—	3	(3)
35	1	1	—	—	2	
36*	2	3	4	6	15	
37	—	2	—	—	2	(2)
38	1	1	2	4	8	
Section B Total	7	16	11	16	50	(7)
Paper Total	20	25	19	16	80	(22)